

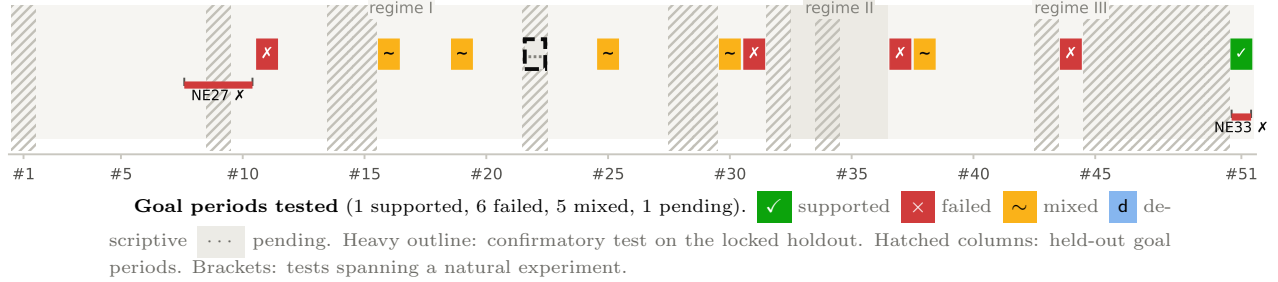
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 1.5/5

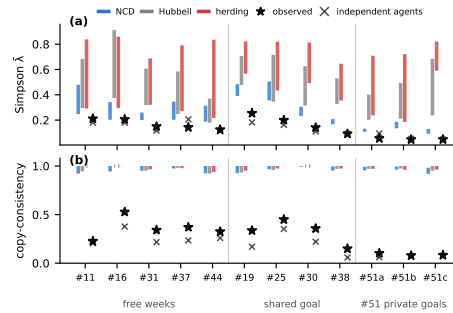
Bottom line. Free LLM swarms don't form a cooperator core: projects are mostly private, so no exchangeable copying model (cooperative, neutral or herding) fits, and copying is only a small excess over independent agents.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Agent slots carry a project label per 30-min window: a cluster of the agent's self-written goal (6 clusterings), or the repo it links. Exact finite- N simulations with the real label gaps fit neutral cooperative dynamics (NCD), Hubbell drift and a herding rival to 9 statistics in 5 free weeks, 4 shared-goal weeks and 3 blocks of #51. Synthetic power: ≈ 0.9 at $N \geq 12$. **NCD failed:** 4/5 free weeks failed, 1 mixed. All three models are rejected (joint check $p < 0.01$ in 27/30 free-week fits). Projects are fragmented: 79–91% are held by one agent (NCD: 45–68%), only 23–53% of switches join a project someone else holds (models: $\approx 100\%$), and Simpson λ falls below every model's range. Copying adds little over independent agents (λ 0.15 vs 0.12 in #31). Frequency dependence has no consistent sign. Batch joins: failed. Holdout #22 frozen, not run.



Observables. Per period, goal clusters (km24). (a) Observed Simpson λ (star) against the 95% predictive ranges of the three fitted copying models (bars), and an independent-agents null (cross; each agent's day sequence shifted). The observation sits below all three models and close to the null. (b) Share of project switches that join a project another agent currently holds. Every copying model predicts ≈ 1 ; free and shared weeks give 0.15–0.53, and #51 (private roles) 0.08–0.10.

PHYSICAL MODEL

Model 06 (Piñero et al. 2025). N individuals (agent slots), each holding one species (project). Per step, with probability μ a new project replaces a random slot C . Otherwise C copies slot A 's project only if A and a third slot B hold different projects: cooperation, so rare projects recruit more per capita. $\lambda = \sum_i x_i^2$ is the Simpson index of shares $x_i = n_i/N$. Rivals: Hubbell (always copy) and herding (copy favoured when A, B agree).

$$b_n = (1 - \mu) \frac{n}{N} \left(1 - \frac{n}{N}\right)^2, \quad P_n \propto \frac{(1 - \mu)^n}{n} e^{-(n - N\lambda^*)^2 / 2N}$$

$$\frac{1}{\mu} = \sqrt{\frac{\pi N}{2}} \operatorname{erfcx}\left(\sqrt{\frac{N}{2}} (\mu - \lambda^*)\right), \quad \mu_B = \frac{e^{-2}}{\sqrt{2\pi N}}, \quad \mu_L = \sqrt{\frac{2}{\pi N}}$$

$$P(\text{recruit to } i) \propto x_i e^{\beta x_i} : \quad \beta < 0 \text{ NCD}, \quad 0 \text{ Hubbell}, \quad > 0 \text{ herding}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Neutral cooperative dynamics doesn't apply to these swarms, and neither does any model where agents are interchangeable copiers. Left to choose, LLM agents mostly run their own projects. Herding (H11) is a small layer on top: agents look at the same repos, but their stated plans stay private. The practical content is modest. If an operator wants a shared project, it has to be assigned; free time won't produce a "core". Two cheap diagnostics fall out: the singleton fraction and copy-consistency, which separate a collaborating swarm from a set of soloists (#51 is the extreme). The elegant theory (bimodal core, λ^* law) never came into range: μ is too high at $N \leq 30$.

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 NCD beats rivals (free weeks)	4 failed, 1 mixed; NCD failed wins 1/30 fits	
Adequacy (joint PPC)	all 3 models fail in 27/30 failed fits (p at floor)	
P3 $\bar{\lambda}$ at NCD value	below every model; singletons 0.79–0.91	failed
P6 shared-goal herding	herding beats NCD 4/4, mixed but $\bar{\lambda}$ below range	
P8 copying > independent	small excess (#31: 0.149 vs 0.117)	failed (rule)
P4 cooperator core	$\hat{\mu}$ never below μ_B	n/a
NE27/NE33 predict-after	every model over-predicts post-join $\bar{\lambda}$	failed

GOAL PERIODS IN DETAIL

G22	...	pending (not run; locked holdout) (holdout)
G11	×	failed (P1; P2 failed; P3 failed; art n/a; no model adequate (joint PPC $p \leq 0.01$ for all thr...
G16	~	mixed (P1; P2 failed; P3 supported; art n/a; no model adequate (joint PPC $p \leq 0.01$ for all t...
G19	~	mixed (P6 contrast)
G25	~	mixed (P6 contrast)
G30	~	mixed (P6 contrast)
G31	×	failed (P1; P2 supported; P3 failed; art failed; no model adequate (joint PPC $p \leq 0.01$ for a...
G37	×	failed (P1; P2 failed; P3 failed; art mixed; no model adequate (joint PPC $p \leq 0.01$ for all t...
G38	~	mixed (P6 contrast)
G44	×	failed (P1; P2 failed; P3 failed; art failed; no model adequate (joint PPC $p \leq 0.01$ for all ...
G51	✓	supported (P7 check passes: no block 'supported'; but the free weeks fail the same way, so t...
NE27	×	failed ((a) failed: post λ outside every model's 95% interval; (b) failed: the three kernel...
NE33	×	failed ((a) failed: post λ outside every model's 95% interval; (b) n/a, j 3 multi-option ne...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

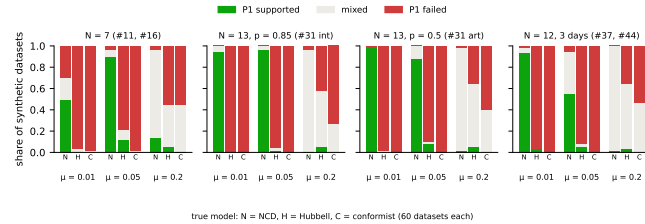
Goal clusters are topics, not repositories, which may inflate singletons; artifact labels are fragmented too. #37 and #38 pool two rooms. With all models rejected, “best by likelihood” tracks predictive spread, so herding’s 20/30 wins are not evidence for herding. Results hold on the shared deterministic project labels (99.4–99.8% agreement). The private-projects result sits against H11’s herding in attention (artifact mentions); the work-output ledger should decide which describes work.

ROUND 2

Agent-specific innovation fields in NCD and Hubbell; repository-anchored species; per-room #37/#38; embedding swap; #22 confirmation after sign-off.

No round-2 redirects yet.

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Second figure. Synthetic power of the P1 rule with exact finite- N simulators on the real label mask: NCD is recovered 0.50–0.98 at $\mu \leq 0.05$, and rivals are falsely supported ≤ 0.12 .